

CLOUD READINESS & MIGRATION **ASSESSMENT REPORT**

Project 1 — Cloud Readiness & Migration Assessment

Field	Details
Project Title	Enterprise Cloud Transformation & AWS Migration Strategy
Project Name	Project 1 — Cloud Readiness & Migration Assessment
Submitted By	Aditya Chandra
Assessment Type	Cloud Readiness & Migration Assessment
Target Architecture	AWS Cloud-Based Modernized Infrastructure
Academic Domain	Cloud Computing & Enterprise Infrastructure
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1. Executive Summary

This report presents a Cloud Readiness and Migration Assessment for ShopEase Pvt Ltd, a mid-sized e-commerce organization currently operating on a traditional on-premises infrastructure. As the company continues to grow, its existing infrastructure faces challenges related to scalability, operational efficiency, maintenance, security, and disaster recovery. Manual infrastructure management, limited monitoring capabilities, and slow deployment processes have created a need for a more scalable and modern technology platform.

The objective of this assessment is to evaluate the organization's readiness for cloud adoption and recommend a secure, scalable, and cost-efficient migration strategy. The report includes cloud readiness evaluation, technical and security gap analysis, application dependency mapping, cloud maturity assessment, TCO vs ROI analysis, AWS target architecture recommendations, and a migration roadmap.

After evaluating the organization's infrastructure and business requirements, Amazon Web Services (AWS) was identified as the most suitable cloud platform due to its scalability, security, operational flexibility, and enterprise-grade cloud services. The proposed migration strategy is expected to improve application availability, scalability, disaster recovery, operational efficiency, and cost optimization while supporting future cloud-native modernization initiatives.

By adopting cloud technologies, automation, Infrastructure as Code, and DevOps practices, ShopEase Pvt Ltd can modernize its IT infrastructure and build a reliable, secure, and scalable platform for future business growth.

2. Organization Overview

Company Background

ShopEase Pvt Ltd is a mid-sized e-commerce company that provides online retail services across categories such as electronics, fashion, lifestyle products, and home appliances. The organization serves a growing customer base through its digital platform and relies heavily on its IT infrastructure for daily operations and customer transactions.

The company currently operates on a traditional on-premises infrastructure hosted in a private data center. With increasing business growth and customer demand, the existing infrastructure has become difficult to scale and manage efficiently.

Existing IT Environment

The current IT infrastructure consists of:

- Web servers hosting the customer-facing portal
- Application servers running monolithic applications
- MySQL databases on physical servers
- NAS-based storage systems
- Traditional firewall appliances
- Manual deployment and backup processes

Most infrastructure operations are managed manually, resulting in limited automation, reduced flexibility, and increased operational overhead.

Current Business Challenges

The organization currently faces several infrastructure and operational challenges, including:

- Infrastructure scaling limitations during peak sales periods
- High operational and maintenance costs
- Slow provisioning and deployment processes
- Downtime risks during maintenance activities
- Limited automation and monitoring capabilities
- Weak disaster recovery mechanisms
- Increasing infrastructure management complexity

To address these challenges, ShopEase Pvt Ltd plans to modernize its infrastructure by adopting scalable and secure cloud technologies.

3. Project Objectives

The primary objectives of this project are:

- Evaluate the organization's readiness for cloud migration
- Identify technical, business, and security gaps
- Analyze application and infrastructure dependencies
- Compare on-premises and cloud financial models
- Measure cloud maturity level
- Recommend the most suitable cloud platform
- Design a migration roadmap
- Recommend remediation and modernization strategies
- Improve scalability, reliability, and security
- Reduce operational and infrastructure costs

4. Existing On-Premises Infrastructure Assessment

Existing Infrastructure Components

Component	Existing Setup
Web Servers	Nginx-based web servers
Application Layer	Java/.NET monolithic applications
Database	MySQL hosted on physical servers
Storage	NAS-based file storage

Backup	Manual on-premises backup server
Security	Traditional firewall appliances
Monitoring	Basic server monitoring
Deployment	Manual deployment processes

Existing Infrastructure Cost

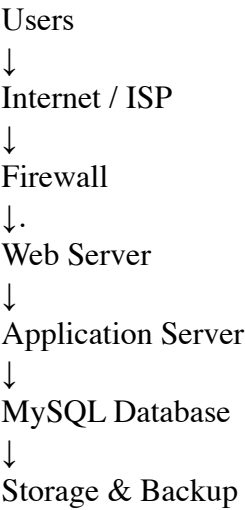
Cost Category	Estimated Cost
Initial Infrastructure Setup	₹2.5 – ₹3.5 Crore
Annual Operational Cost	₹1 – ₹1.5 Crore
Hardware Maintenance	₹40 – ₹60 Lakhs/year
Software Licensing	₹20 – ₹30 Lakhs/year

Existing Infrastructure Challenges

Challenge	Impact
Limited scalability	Poor performance during peak traffic
Manual deployment process	Slow release cycles
Weak disaster recovery	Increased downtime risk
Limited monitoring	Delayed issue detection
High infrastructure cost	Increased operational expenditure
Hardware dependency	Reduced operational flexibility

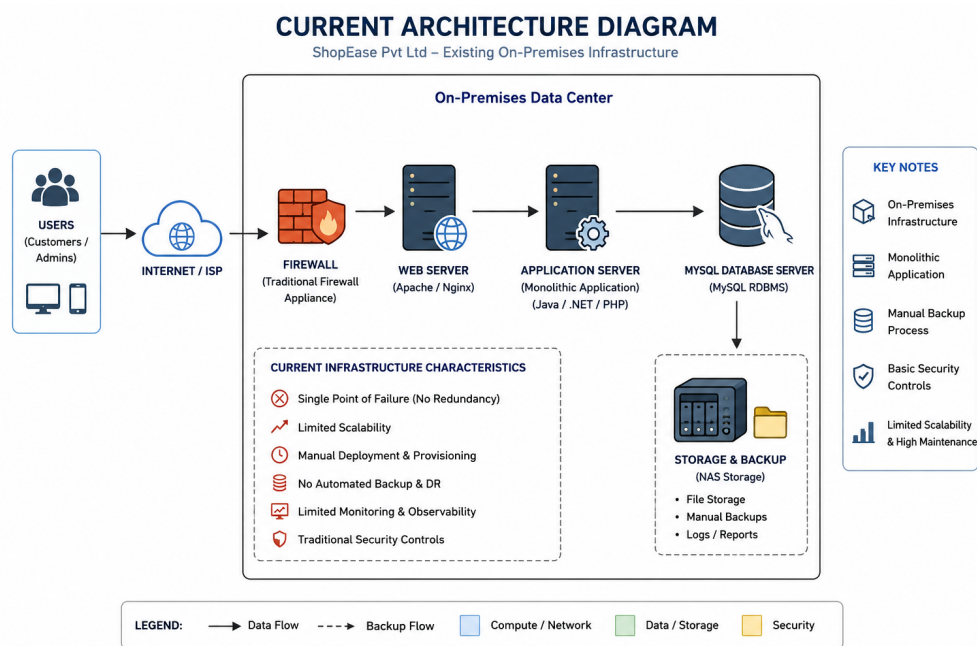
5. Current Architecture Overview

Existing Architecture Flow



Current Infrastructure Characteristic

Area	Current State
Scalability	Limited
Availability	Moderate
Automation	Minimal
Security	Traditional perimeter-based
Monitoring	Basic monitoring only
Disaster Recovery	Weak
Deployment Speed	Slow
Operational Agility	Limited



6. Cloud Readiness Assessment

Cloud readiness assessment evaluates whether the organization is prepared for successful cloud adoption.

Assessment Areas

- Infrastructure readiness
- Application readiness
- Security readiness
- Operational readiness
- Governance readiness
- DevOps maturity
- Organizational skills

Readiness Assessment Scores

Assessment Area	Score (Out of 10)
Infrastructure Readiness	4
Application Readiness	3
Security Readiness	3
DevOps Readiness	2
Operational Readiness	3
Governance Readiness	2
Organizational Skills	3

Overall Readiness Score

3.0 / 10

Readiness Level

Low to Moderate Cloud Readiness

Key Findings

- Existing applications are tightly coupled
- Limited automation capabilities exist
- Security controls require modernization
- Disaster recovery mechanisms are weak
- Cloud-native practices are not implemented
- Operational monitoring is limited

7. Technical Gap Analysis

Technical Gap 1: Monolithic Application Architecture

Current Situation

The organization currently uses a traditional monolithic application architecture where all application components are tightly integrated into a single system.

Problem Identified

This architecture creates scalability limitations and makes application maintenance difficult. Even small changes require rebuilding and redeploying the entire application.

Recommended Solution

The organization should adopt a microservices-based cloud-native architecture. Breaking the application into independent services will improve scalability, maintainability, and deployment flexibility.

Technical Gap 2: Fixed On-Premises Infrastructure

Current Situation

The company relies on traditional on-premises servers with fixed hardware capacity.

Problem Identified

The infrastructure lacks dynamic auto-scaling capabilities, making it difficult to handle fluctuating workloads efficiently.

Recommended Solution

Cloud-based auto-scaling services should be implemented to dynamically increase or decrease resources based on workload demand.

Technical Gap 3: Manual Deployment and Provisioning

Current Situation

Infrastructure provisioning and application deployment processes are currently performed manually.

Problem Identified

Manual operations increase operational complexity and deployment delays while introducing human errors.

Recommended Solution

The organization should implement Infrastructure as Code (IaC) and CI/CD pipelines to automate deployments and infrastructure provisioning.

Technical Gap 4: Limited Monitoring and Logging

Current Situation

The current infrastructure uses basic monitoring tools with limited visibility into system performance and application health.

Problem Identified

There is no centralized observability platform for monitoring logs, metrics, and system events.

Recommended Solution

Centralized monitoring and logging tools should be implemented to improve visibility and proactive issue detection.

Technical Gap 5: Legacy Database Deployment

Current Situation

The database is hosted on a single on-premises server.

Problem Identified

The current deployment creates a single point of failure with limited high availability capabilities.

Recommended Solution

The organization should migrate to managed multi-availability-zone cloud databases with automated failover and backup mechanisms.

Technical Gap 6: Tight Coupling Between Services

Current Situation

Application services are tightly coupled with backend databases and internal components.

Problem Identified

This tight integration reduces flexibility and complicates migration or modernization efforts.

Recommended Solution

Services should be decoupled using APIs, containers, and service-oriented design principles.

Technical Gap 7: Weak Backup and Disaster Recovery

Current Situation

The existing backup mechanisms are limited and primarily manual.

Problem Identified

The organization lacks a robust disaster recovery strategy and automated backup solutions.

Recommended Solution

Automated backup and cloud-based disaster recovery solutions should be implemented to ensure business continuity.

8 Business Gap Analysis

Business Gap 1: Dependency on Physical Infrastructure

Current Situation

The organization heavily depends on physical data center infrastructure for its operations.

Problem Identified

This dependency limits flexibility and slows down infrastructure expansion.

Recommended Solution

Migrating to scalable cloud infrastructure will improve agility and support rapid business growth.

Business Gap 2: High Capital Expenditure

Current Situation

The company invests heavily in purchasing and maintaining physical hardware.

Problem Identified

The current model results in high upfront capital expenses and inefficient resource utilization.

Recommended Solution

A pay-as-you-go cloud pricing model should be adopted to optimize infrastructure costs.

Business Gap 3: Manual Operational Processes

Current Situation

Many operational workflows are handled manually.

Problem Identified

Manual processes reduce productivity and increase administrative workload.

Recommended Solution

Workflow automation and cloud-based deployment pipelines should be implemented.

Business Gap 4: Limited Scalability for Business Growth

Current Situation

The current infrastructure struggles to support rapidly increasing customer demand.

Problem Identified

Scalability limitations restrict the organization's ability to grow efficiently.

Recommended Solution

Elastic cloud infrastructure should be implemented to support scalable workloads.

Business Gap 5: Downtime During Maintenance**Current Situation**

System maintenance activities often require service interruptions.

Problem Identified

The infrastructure lacks high-availability capabilities.

Recommended Solution

Highly available cloud architecture with redundancy and failover mechanisms should be adopted.

Business Gap 6: Slow Environment Provisioning**Current Situation**

Provisioning new environments takes significant time and manual effort.

Problem Identified

Development and testing environments cannot be created quickly.

Recommended Solution

Self-service provisioning using cloud platforms and automation tools should be enabled.

9. Security Gap Analysis**Security Gap 1: Basic Security Controls****Current Situation**

The organization primarily relies on traditional perimeter-based security mechanisms.

Problem Identified

The existing security model lacks advanced cloud-native security capabilities.

Recommended Solution

A zero-trust security architecture should be implemented with layered security controls.

Security Gap 2: Weak Identity and Access Management

Current Situation

Access control mechanisms are limited and inconsistently managed.

Problem Identified

Users may have excessive privileges beyond their operational requirements.

Recommended Solution

Role-Based Access Control (RBAC) and cloud IAM policies should be implemented.

Security Gap 3: Inconsistent Patch Management

Current Situation

System and software patching processes are performed manually and irregularly.

Problem Identified

Unpatched systems create security vulnerabilities.

Recommended Solution

Automated patch management solutions should be implemented.

Security Gap 4: Lack of Centralized Security Monitoring

Current Situation

Security monitoring is fragmented across multiple systems.

Problem Identified

There is no centralized visibility into security events and threats.

Recommended Solution

SIEM and centralized security monitoring solutions should be deployed.

Security Gap 5: Weak Disaster Recovery Preparedness

Current Situation

The organization has limited disaster recovery planning.

Problem Identified

Recovery capabilities are insufficient for major infrastructure failures.

Recommended Solution

A cloud-based disaster recovery strategy should be established.

Security Gap 6: Limited Encryption Mechanisms

Current Situation

Encryption is not consistently implemented across systems and data flows.

Problem Identified

Sensitive data may remain exposed during storage or transmission.

Recommended Solution

Encryption at rest and in transit should be enforced across all systems.

Security Gap 7: Manual Compliance Management

Current Situation

Compliance activities are handled manually.

Problem Identified

Manual compliance management increases administrative effort and audit complexity.

Recommended Solution

Automated compliance and governance tools should be implemented.

10. Application Dependency Mapping

Internal Dependency Flow

```

Users
↓
Frontend Application
↓
Application Server
↓
MySQL Database
↓
Storage Layer

```

External Dependencies

External Service	Purpose
Payment Gateway APIs	Online payment processing
SMTP/SMS Services	Notifications and alerts
Logistics APIs	Order tracking

Dependency Risks

- Tight application coupling
- Single database dependency
- Limited service isolation
- Manual integration management

11. Cloud Maturity Scoring Sheet

Cloud Maturity Assessment

Domain	Maturity Level
Infrastructure	Level 2
Security	Level 2
Automation	Level 1
Governance	Level 1
DevOps	Level 1
Monitoring	Level 2

Overall Cloud Maturity Score

1.7 / 5

Current Maturity Level

Level 2 — Opportunistic / Virtualized Environment

Interpretation

The organization currently has:

- Basic virtualization capabilities
- Limited automation practices
- Minimal cloud-native adoption

- Weak governance controls
- Limited DevOps implementation

The organization lacks:

- Infrastructure as Code
- Advanced governance controls
- Automated security mechanisms
- Mature operational processes

12. TCO vs ROI Comparison

3-Year Cost Comparison

Environment	Estimated Cost
On-Premises Infrastructure	₹7 – ₹9 Crore
AWS Cloud Infrastructure	₹2 – ₹3 Crore

Expected Benefits After Migration

Benefit	Expected Improvement
Infrastructure Cost Reduction	30–40%
Availability	99.99% uptime
Deployment Speed	Weekly to daily
Recovery Time	Significant reduction
Scalability	On-demand scaling

ROI Formula

$$ROI = (Benefits - Cost) / Cost \times 100$$

Expected ROI

The organization is expected to achieve positive ROI within 18–24 months after migration due to reduced infrastructure expenditure and improved operational efficiency.

Why AWS Was Selected

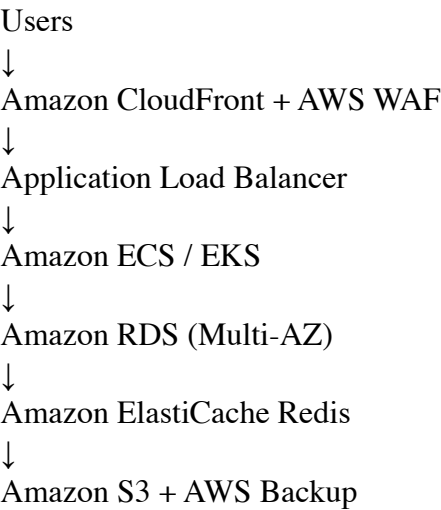
AWS was selected because of:

- Extensive service ecosystem
- Mature migration services
- Enterprise-grade scalability
- Advanced networking and security

- Strong DevOps ecosystem
- Global infrastructure presence
- Better operational flexibility

13. Proposed AWS Target Architecture

Proposed AWS Architecture Flow



Recommended AWS Services

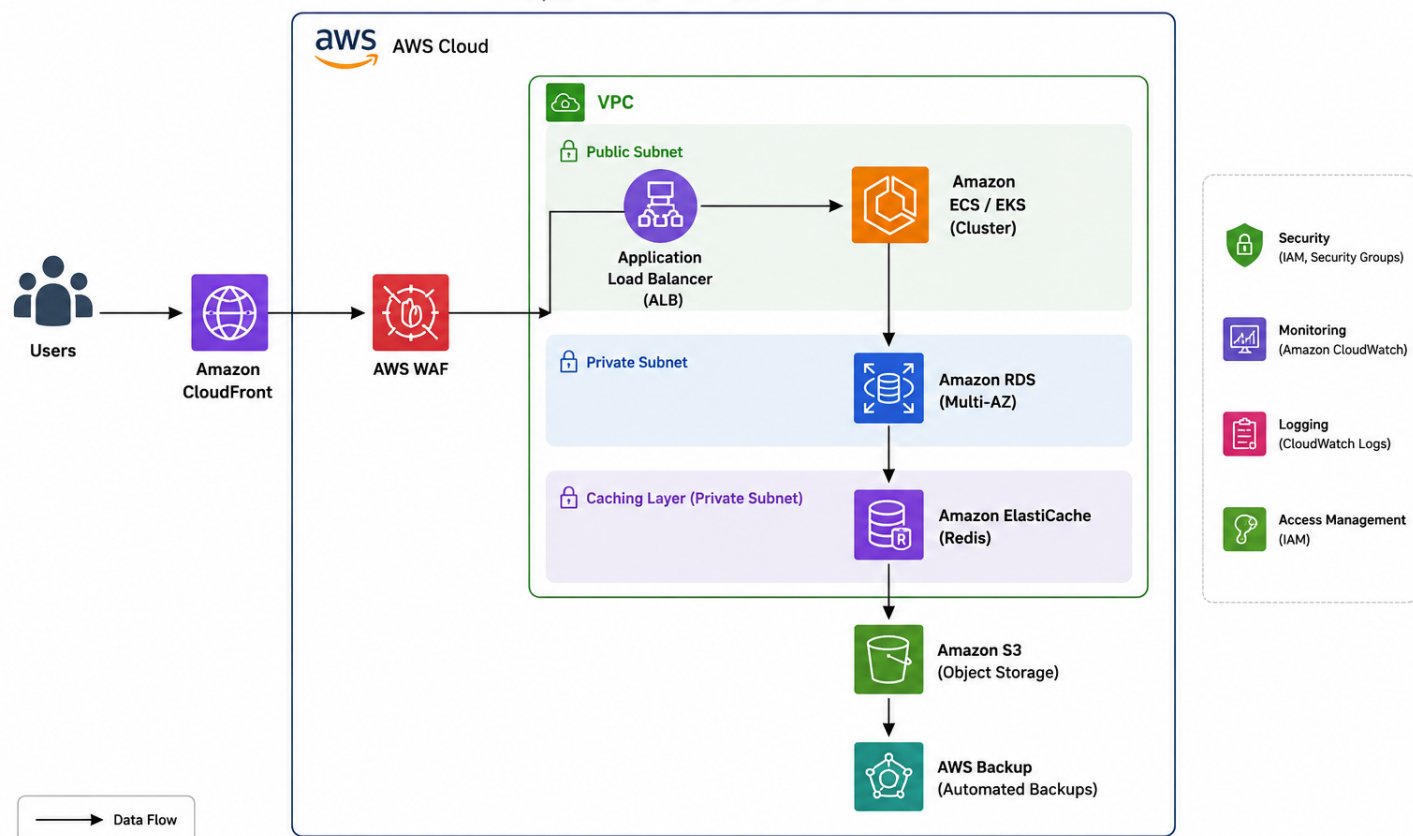
AWS Service	Purpose
Amazon ECS / EKS	Container orchestration
Amazon RDS	Managed relational database
Amazon S3	Object storage
AWS WAF	Web application firewall
Amazon CloudFront	Content delivery network
Amazon CloudWatch	Monitoring and logging
AWS IAM	Identity and access management
AWS Backup	Backup management

Benefits of Proposed Architecture

- High availability
- Improved scalability
- Reduced operational overhead
- Improved security posture
- Better disaster recovery capabilities
- Faster application deployment

PROPOSED AWS ARCHITECTURE FLOW

ShopEase Pvt Ltd – Future State on AWS



14. Migration Strategy (6Rs Model)

6R Migration Strategies

Strategy	Description
Rehost	Lift-and-shift migration
Replatform	Minor cloud optimization
Refactor	Cloud-native modernization
Retire	Remove unused applications
Retain	Keep workloads on-prem temporarily
Repurchase	Replace with SaaS solution

Recommended Migration Strategy

Workload	Migration Approach
Web Applications	Replatform
Databases	Rehost/Replatform
Legacy Systems	Retain initially
Monitoring Tools	Replace with cloud-native tools

15. Migration Roadmap & Timeline

Phase 1 — Assessment (2 Weeks)

Activities:

- Infrastructure analysis
- Dependency mapping
- Security assessment
- Cost analysis

Phase 2 — Planning (3 Weeks)

Activities:

- AWS environment setup
- IAM configuration
- Network architecture planning
- Security policy definition

Phase 3 — Migration (2–3 Months)

Activities:

- Application migration
- Database migration
- Network configuration
- Security validation

Phase 4 — Validation (2 Weeks)

Activities:

- Performance testing
- Security testing
- Disaster recovery testing
- Failover validation

Phase 5 — Optimization (Ongoing)

Activities:

- Cost optimization
- Auto scaling improvements
- Monitoring enhancements
- Operational optimization

16. Remediation & Modernization Plan

Infrastructure Modernization

- Adopt containerized workloads
- Implement Kubernetes orchestration
- Use managed database services
- Implement auto scaling
- Introduce Infrastructure as Code

Security Modernization

- Implement centralized IAM
- Enable Multi-Factor Authentication
- Introduce security monitoring
- Implement encryption across workloads
- Enable automated compliance checks

Operational Modernization

- Introduce CI/CD practices
- Improve monitoring and logging

- Implement automated backups
- Adopt DevOps methodologies
- Improve operational governance

17. Final Recommendations

Primary Recommendation

Migrate ShopEase Pvt Ltd to AWS using a phased hybrid migration strategy.

Key Recommendations

- Adopt AWS as the primary cloud provider
- Modernize applications gradually
- Implement Infrastructure as Code
- Improve governance and security controls
- Introduce CI/CD automation
- Enable centralized monitoring
- Use Multi-AZ deployment for high availability
- Train teams on cloud-native operations

18. Conclusion

The cloud readiness assessment shows that ShopEase Pvt Ltd currently operates in a low-to-moderate cloud maturity environment with challenges related to scalability, automation, security, and disaster recovery. The existing on-premises infrastructure creates operational complexity, higher maintenance costs, and limited flexibility for future growth.

Migrating to AWS will help the organization improve scalability, reliability, security, operational efficiency, and cost optimization while supporting business growth through a more modern and resilient cloud environment. By adopting cloud-native technologies, automation, Infrastructure as Code, and DevOps practices, ShopEase Pvt Ltd can successfully modernize its IT infrastructure and build a secure, scalable, and future-ready platform.

